# Malaria Sporozoite Detection by Dissection and ELISA to Assess Infectivity of Afrotropical *Anopheles* (Diptera: Culicidae)

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**ABSTRACT** Malaria infection rates determined by dissection and *Plasmodium falciparum* enzyme-linked immunosorbent assay (ELISA) were compared for 26,935 *Anopheles gambiae* Giles sensu lato and 17,739 *Anopheles funestus* Giles collected during 20 mo in western Kenya. ELISA infection rates were about 43% higher than dissection sporozoite rates. In dissection-negative *Anopheles*, circumsporozoite (CS) protein was detected by ELISA in 5.2% of 10,017 salivary gland samples and in 12.2% of 237 thorax samples. The accuracy of dissection and ELISA techniques was compared by the following tests on a group of 352 field-collected *Anopheles* (held 10 d to ensure sporogonic development): salivary gland dissection, examination of Giemsa-stained dissection slides, ELISA tests on salivary gland and thorax body parts, and microscopic techniques for determining sporozoite loads. Respective infection rates were 9.9%, 10.8%, and 15.6% for dissection, stained slides, and ELISA. Sporozoite loads were associated significantly with ELISA absorbance values ($r = 0.76$). Compared with Giemsa-stained dissection slide results, the sensitivity of sporozoite detection was 92.1% for dissection compared with 78.9% for ELISA; specificity was 100.0% for dissection versus 92.0% for ELISA. Immunological detection of CS protein in head–thorax samples of Afrotropical vectors overestimated the proportion of infective *Anopheles* because the comparison of techniques indicated that 45.4% of the ELISA positive *Anopheles* did not contain salivary gland sporozoites.

**KEY WORDS** Insecta, *Anopheles* spp., malaria, ELISA

IMMUNOLOGICAL TECHNIQUES for detecting sporozoite infections in mosquitoes have enhanced the potential for measuring malaria infection rates in *Anopheles* populations. Following the initial development of immunoradiometric assays (IRMA) using *Plasmodium* specific monoclonal antibodies for sporozoite detection (Zavala et al. 1982, Collins et al. 1985), enzyme-linked immunosorbent assays (ELISA) were developed and standardized for *P. falciparum* (Burkot et al. 1984, Wirtz et al. 1987b), *P. vivax* (Wirtz et al. 1985), and recently, for *P. malariae* and *P. ovale* (M. S. Beier et al. 1988, Collins et al. 1988). These techniques offer several theoretical advantages over standard dissection methods (Zavala et al. 1982, Burkot et al. 1984, Collins et al. 1985, Wirtz et al. 1985). IRMA and ELISA techniques have been evaluated for field use (Collins et al. 1984, J. C. Beier et al. 1987, Wirtz et al. 1987a, Boudin et al. 1988) and, in many malaria field programs, have replaced dissection as the method of choice for determining sporozoite infection rates.

Sporozoite rates by dissection measure the proportion of female *Anopheles* with sporozoites in the salivary glands, a direct measure of potential infectivity (Macdonald 1952). Positive reactions in the monoclonal antibody-based assays measure the presence of *Plasmodium* circumsporozoite (CS) protein originating from sporozoites or from developing oocysts on the midgut. Positive reactions caused by oocysts, for whole mosquitoes tested by ELISA or IRMA, can yield infection rates of questionable epidemiological value. A closer agreement with dissection sporozoite rates can be attained by cutting mosquitoes at the thorax and testing anterior portions (J. C. Beier et al. 1987). Similarly, sporozoite rates by dissection can be underestimated by dissection error. Dissection and immunological techniques are useful for measuring relative, temporal changes in malaria infection rates, but further efforts are needed to determine how accurately each technique actually measures the proportion of mosquitoes containing sporozoites in the salivary glands.

During concurrent use of dissection and *P. falciparum* ELISA techniques in western Kenya over a 20-mo period, ELISA tests on anterior portions of cut mosquitoes consistently yielded higher infection rates than did dissections. The accuracy of dissection and ELISA techniques for actual sporozoite detection in salivary glands was determined

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by relating the presence of circumsporozoite protein in salivary gland and thorax samples to the presence or absence of counted sporozoites in the salivary glands.

## Materials and Methods

***Anopheles* Collections.** *Anopheles* were collected by aspiration inside houses from November 1985 to June 1987 near Kisumu in western Kenya. Weekly collections were made from >50 houses in the village of Kisian (10 km west of Kisumu) and in Saradidi (55 km west of Kisumu). *An. gambiae* Giles sensu lato and *An. funestus* Giles constituted >99% of the collected *Anopheles* and were the only species tested in this study. Study sites, collection methods, mosquito identification and handling procedures are described by J. C. Beier et al. (1987). Malaria infections in the human population in this area include 87.1% *P. falciparum*, 1.6% *P. malariae*, 0.4% *P. ovale*, and 11.0% mixed infections involving mainly *P. falciparum* (Spencer et al. 1987). Hyperendemic malaria transmission in these two sites was studied by the World Health Organization from 1972 to 1976 (Fontaine et al. 1978).

**Malaria Infection Rates in *Anopheles*.** Malaria infection rates were determined by dissection and sporozoite ELISA techniques. Up to 100 parous *Anopheles* were dissected each day within 8 h of collection. Mosquitoes were first examined for parity. Parous *Anopheles* were dissected for midgut oocysts and salivary gland sporozoites using standard techniques (World Health Organization 1975). Sporozoite rates were calculated as the number of females with sporozoites divided by the total dissected (parous plus nulliparous).

*Anopheles* for ELISA testing were held at −20°C in Kisumu before transport to Nairobi, where they were held at −70°C. To eliminate the detection of CS antigen from oocysts on the midgut, *Anopheles* were divided into two parts by cutting them transversely at the thorax between the first and third pairs of legs (Beier et al. 1987). The anterior portion of each mosquito was prepared for ELISA testing by grinding in 50 μl blocking buffer (BB) [for 1 liter: Bovine serum albumin (BSA), 10.0 gm; casein, 5.0 gm; thimerosal, 0.1 gm; phenol red, 0.01 gm; 1,000 μl phosphate buffered saline (PBS), pH 7.4] with nonindet P-40 (NP-40) (5 μl NP-40 : 1 ml BB) (Sigma Chemical Company, St. Louis), then adding 200 μl BB to bring the final volume to 250 μl per mosquito sample. Individual females were tested for *P. falciparum* CS antigen using the 2A10 monoclonal antibody according to methods described by Wirtz et al. (1987b). Two positive controls on each plate consisted of 100 and 10 pg of a recombinant *P. falciparum* CS protein ($R32tet_{32}$) (Wirtz et al. 1987a). Negative controls on each plate included four field-collected male *An. gambiae* s.l. or *An. funestus*. Samples were considered positive if absorbance values (range, 0–2.00) determined by an ELISA reader at 30 min after the addition of substrate exceeded twice the mean of the negative controls (J. C. Beier et al. 1988). Monthly dissection and ELISA infection rates for each *Anopheles* species at each study site were compared by the test for equality of percentages (Sokal & Rohlf 1969). The same test was used to compare overall mean infection rates for all months when the two methods were used. Correlation analysis was used to compare relative changes in infection rates measured by the two tests over time.

**ELISA Testing of Dissected Salivary Glands.** Salivary gland material was obtained from dissection slides by removing the coverslip, rinsing the coverslip with about 20 μl physiological saline, and then transferring about 60 μl of gland material from the slide into a 2.0-μl polypropylene vial using a 20-μl capillary tube (J. C. Beier et al. 1987). Following the addition of 100 μl BB, samples were frozen at −20 or −70°C in Nairobi. Before testing by ELISA, 50 μl BB with NP-40 were added to each sample. From 11 August 1986 to 26 February 1987, salivary gland material was tested from all dissected, parous *Anopheles*. Positive gland material then was tested by ELISA to determine the proportion of sporozoite infections that were *P. falciparum*. Negative salivary glands were tested to identify infections existing either as soluble CS protein (Verhave et al. 1988) or as sporozoites possibly missed by dissection.

**ELISA Testing of Thorax Samples.** ELISA tests were done on thorax body parts of dissected, parous *Anopheles* to determine if CS protein could be detected by ELISA after removal of salivary glands. Gland positive and negative *Anopheles* dissected from November 1987 to February 1988 were held dry in individual vials, randomly selected and the thoraces prepared for ELISA testing by removing wings and legs (heads were previously removed during dissection). Each thorax was ground in 50 μl BB-NP40, and 200 μl BB were added; samples were tested by *P. falciparum* ELISA.

**Dissection and ELISA Comparison for *Anopheles* Held 10 d.** From 9 March to 8 April 1988, *Anopheles* were collected on 7 d inside houses by aspiration in the Kisian site. Mosquitoes were placed in screened gallon cartons by species and held at room temperature in Kisumu (18–27°C) for 10 d, to allow sporogonic development of the malaria parasites. The mosquitoes were provided with 10% sucrose solution, and cartons were covered with black cloth, which was moistened daily. A total of 352 parous *Anopheles*, surviving 10 d, were used to evaluate sporozoite detection techniques. As previously described, gut and salivary glands of parous mosquitoes were dissected, and salivary glands and thoraces were saved and tested by ELISA. Dissection slides from all *Anopheles* were held dry for 3 d and then stained with 4% Giemsa stain diluted in phosphate buffer (pH 6.8) for 40 min; slides were not fixed in methanol. For gland positive slides, 5 μl of the 60-μl aliquot for ELISA testing were placed

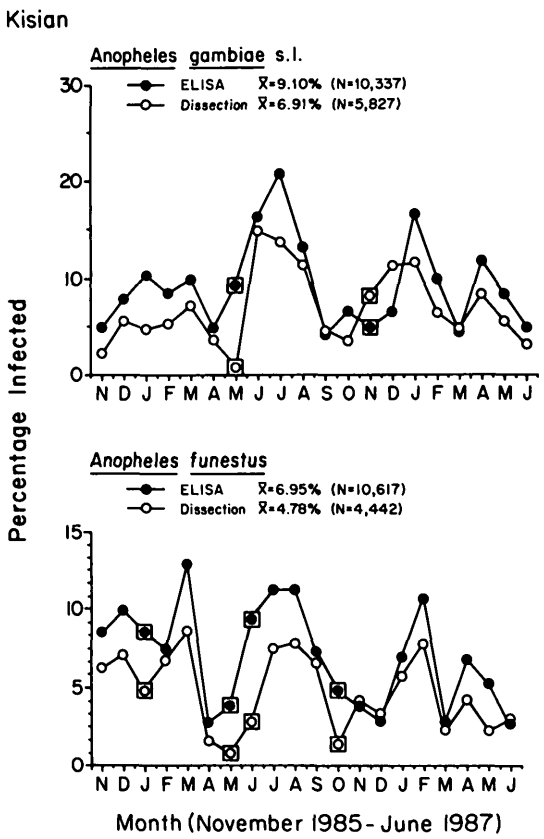



**Fig. 1.** Comparison of dissection and *P. falciparum* ELISA infection rates for *An. gambiae* s.l. and *An. funestus* in the Kisian study site, western Kenya, over 20 mo. Monthly mean infection rates and total sample sizes are shown. For months in which significant differences in infection rates were detected between the two techniques, points are enclosed by a square.

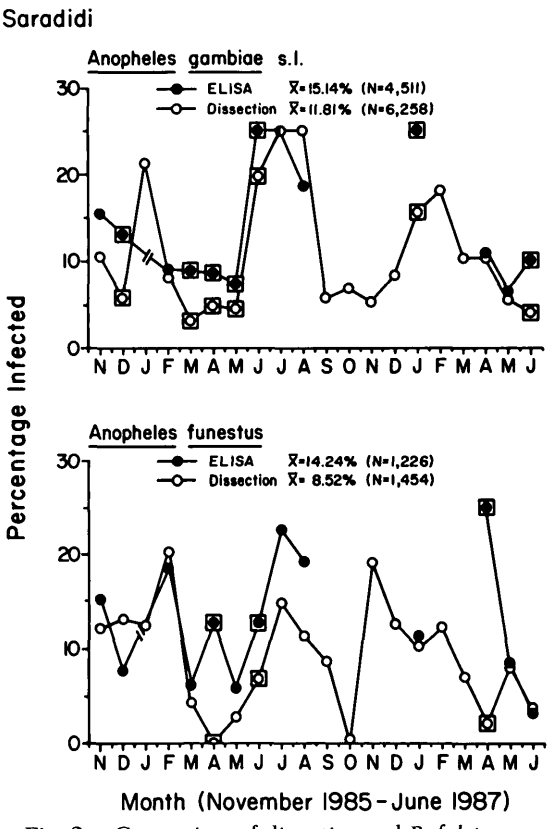



**Fig. 2.** Comparison of dissection and *P. falciparum* ELISA infection rates for *An. gambiae* s.l. and *An. funestus* in the Saradidi study site, western Kenya, over 20 mo. Monthly mean infection rates and total sample sizes are shown. For months in which significant differences in infection rates were detected between the two techniques, points are enclosed by a square.

onto one spot (5 mm diameter) of a standard immunofluorescence slide (Carlson Scientific, Peotone, Ill.); these were held 3 d and stained as above. This spot and the area on the dissection slide containing salivary gland material (circumscribed with a diamond pencil immediately after dissection) were examined for sporozoites after staining by examining every field at 400×. The number of sporozoites per gland positive mosquito was calculated as previously described (J. C. Beier et al. 1989).

## Results

**Dissection and ELISA Infection Rates.** Malaria infection rates determined by dissection and *P. falciparum* ELISA were compared over a 20-mo period for *An. gambiae* s.l. and *An. funestus* in the villages of Kisian (Fig. 1) and Saradidi (Fig. 2), western Kenya. Monthly infection rates determined by dissection and ELISA were correlated positively for *An. gambiae* s.l. ($r = 0.75$, df = 18, $P = 0.0001$) and *An. funestus* ($r = 0.84$, df = 18, $P = 0.0001$) in Kisian, and *An. gambiae* s.l. ($r = 0.86$, df = 11, $P = 0.0002$), but not *An. funestus* ($r = 0.35$, df = 11, $P = 0.246$) in Saradidi. However, ELISA infection rates were consistently higher than dissection sporozoite rates. For *An. gambiae* s.l. in Kisian, the average monthly ELISA infection rate of 9.10% ($n$ = 10,337) was 31.7% higher than the dissection sporozoite rate of 6.91% ($n$ = 5,827) ($t$ = 4.94, df = 38, $P < 0.01$). For *An. funestus*, the monthly average ELISA infection rate (6.95%; $n$ = 10,617) was 45.4% higher than the dissection sporozoite rate (4.78%; $n$ = 4,442) ($t$ = 5.19, df = 26; $P < 0.01$). Over 14 mo for *An. gambiae* s.l. in Saradidi, the average monthly ELISA infection rate (15.14%; $n$ = 4,511) was 28.2% higher than by dissection (11.81%; $n$ = 6,258) ($t$ = 5.00, df = 26, $P < 0.001$). ELISA infection rates for *An. funestus* in Saradidi (14.24%; $n$ = 1,226) were 67.1% higher than dissection (8.52%; $n$ = 1,454) over 13 mo ($t$ = 4.68, df = 24, $P < 0.01$).

**ELISA Testing of Salivary Gland Samples.** *P. falciparum* CS protein was detected by ELISA in 5.2% (526/10,017) of salivary gland samples considered negative by dissection (Table 1). Oocysts on the midgut were detected by dissection in 2.5%

**Table 1. Detection of *P. falciparum* CS protein by ELISA in salivary gland samples from *Anopheles*, which were found to be negative or positive for sporozoites by dissection**

| Dissection result | Salivary glands tested by ELISA | | | |
|---|---|---|---|---|
| | *An. gambiae* s.l. | | *An. funestus* | |
| | No. tested | % Positive | No. tested | % Positive |
| Negative | 7,591 | 5.2 | 2,426 | 5.3 |
| Positive | 632 | 87.0 | 183 | 82.5 |

**Table 2. Detection of *P. falciparum* CS protein by ELISA in thoraces from dissection positive or negative *Anopheles*, after removal of salivary glands**

| | Sporozoite result from dissection | No. tested | % Positive by ELISA | |
|---|---|---|---|---|
| | | | Salivary gland | Thorax |
| *An. gambiae* s.l. | Positive | 126 | 88.1 | 77.0 |
| | Negative | 198 | ND | 9.1 |
| *An. funestus* | Positive | 17 | 76.5 | 88.2 |
| | Negative | 39 | ND | 28.2 |

N.D., not determined.

of the 10,017 *Anopheles* with negative salivary glands; 42.8% (107/250) of these oocyst infections were positive by *P. falciparum* ELISA. Salivary gland samples considered negative by dissection were ELISA positive in 10.8% (27/250) of the *Anopheles* with oocysts, and in 15.9% (17/107) when oocysts were *P. falciparum* positive by ELISA. ELISA detection of CS protein in the salivary glands was not associated significantly with the presence of oocysts.

**ELISA Testing of Thorax Samples.** Thorax samples from 143 dissection positive and 237 dissection negative *Anopheles* were tested by *P. falciparum* ELISA (Table 2). After salivary glands were removed by dissection, 78.9% (112/142) of the thorax samples from positive dissections were correspondingly positive by ELISA; this was not significantly different from the 87.3% (124/142) rate of ELISA detection of sporozoites in salivary gland samples. ELISA detected *P. falciparum* CS protein in 12.2% (29/237) of the thorax samples from gland negative *Anopheles*; the rate of CS protein detection in *An. funestus* (28.2%) was higher than for *An. gambiae* s.l. (9.1%) ($\chi^2 = 11.83$, df = 1, $P < 0.01$). Overall, absorbance values for ELISA positive thorax samples averaged 1.09 ± 0.62 ($n = 112$) when sporozoites were detected in glands by dissection compared with 0.41 ± 0.49 ($n = 29$) when salivary glands were negative by dissection ($F = 29.00$; df = 1, 139; $P = 0.0001$).

**Dissection and ELISA Comparison for *Anopheles* Held 10 d.** Sporozoites were detected by dissection in 9.9% (35/352) of *Anopheles* held 10 d after capture. When dissection slides were stained by Giemsa stain and reexamined, sporozoites were detected in 100% (35/35) of the slides from dissection positive *Anopheles* and in three considered negative. Similar sporozoite rates were observed in *An. gambiae* s.l. (10.9%; $n = 221$) and *An. funestus* (10.7%; $n = 131$).

Salivary glands and thoraces from dissected *Anopheles* were tested by ELISA and results were compared with sporozoite detection in Giemsa-stained slides (Table 3). For the 38 *Anopheles* with sporozoites found in salivary glands, ELISA detected CS protein in 79% of gland samples and in 65% of thorax samples; all positive thorax samples had corresponding CS protein in gland samples. For 314 slide negative *Anopheles*, CS protein was detected in 6.4% of gland samples and in 5.1% of thorax samples. ELISA positive thorax samples were detected in four *Anopheles* with negative glands as determined by dissection and ELISA; ELISA absorbance values in these females averaged 0.70 ± 0.38.

ELISA results and absorbance values for salivary gland and thorax samples were examined in relation to the number of sporozoites detected from salivary glands (Table 4). For ELISA positive salivary gland samples, absorbance values for slide positive samples were significantly higher than for slide negative samples ($F = 16.16$; df = 1, 48; $P = 0.002$). Absorbance values for thorax samples from slide positive mosquitoes were not significantly different from those *Anopheles* where no sporozoites were detected.

The number of sporozoites from slides was associated positively with ELISA absorbance values for salivary gland ($r = 0.76$, df = 36, $P = 0.0001$) and thorax ($r = 0.48$, df = 36, $P = 0.002$) samples. As seen in Table 4, 96.7% (29/30) of gland and 52.1% (13/25) of thorax samples from slide positive mosquitoes yielded absorbance values >0.50. For slide negative mosquitoes, absorbance values were over 0.50 for 65.0% (13/20) of salivary gland and 56.2% (9/16) of thorax samples. Circumsporozoite protein was not detected by ELISA in salivary glands from eight *Anopheles* containing the following numbers of sporozoites: 2, 3, 3, 9, 10, 12, 418, and 2,460. This group of *Anopheles* contained 79 times fewer sporozoites than those with ELISA positive glands ($F = 26.00$; df = 1, 36; $P = 0.0001$). Similarly, 13 *Anopheles* with ELISA negative thor-

**Table 3. Detection of *P. falciparum* CS protein by ELISA in relation to sporozoite detection in stained dissection slides for *Anopheles* held 10 d after capture**

| | Sporozoite result, stained slides | No. tested | % Positive by ELISA | | |
|---|---|---|---|---|---|
| | | | Salivary gland | Thorax | Salivary gland or thorax |
| *An. gambiae* s.l. | Positive | 24 | 83.3 | 66.7 | 83.3 |
| | Negative | 197 | 7.6 | 5.6 | 9.1 |
| *An. funestus* | Positive | 14 | 71.4 | 64.3 | 71.4 |
| | Negative | 117 | 4.3 | 4.3 | 6.0 |

**Table 4. Comparison of ELISA results and absorbance values for salivary gland and thorax samples in relation to the number of sporozoites detected in Giemsa-stained dissection slides from *Anopheles* held 10 d after capture**

| ELISA sample | Test result | | No. tested | ELISA absorbance | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | Stained slides | ELISA | | $\bar{x} \pm$ SD | <0.50 | | | >0.50 | |
| | | | | | *n* | % | G.M. sporozoites[a] | *n* | % | G.M. sporozoites |
| Glands | Positive | Positive | 30 | 1.45 ± 0.47 | 1 | 3.3 | 72 | 29 | 96.7 | 1,913 |
| | Positive | Negative | 8 | 0.08 ± 0.01 | 8 | 100 | 22 | 0 | 0 | 0 |
| | Negative | Positive | 20 | 0.85 ± 0.57 | 7 | 35.0 | 0 | 13 | 65.0 | 0 |
| Thorax | Positive | Positive | 25 | 0.65 ± 0.57 | 12 | 48.0 | 699 | 13 | 52.0 | 3,539 |
| | Positive | Negative | 13 | 0.08 ± 0.02 | 13 | 100 | 132 | 0 | 0 | 0 |
| | Negative | Positive | 16 | 0.56 ± 0.27 | 7 | 43.8 | 0 | 9 | 56.2 | 0 |

[a] G.M., geometric mean.

aces contained the following numbers of sporozoites: 2, 3, 3, 9, 10, 12, 76, 418, 691, 2,460, 4,799, 7,249, and 31,376; 5 of 7 with ≥76 sporozoites contained ELISA positive glands. These *Anopheles* in which CS protein was not detected by ELISA in the thorax contained about 12 times fewer sporozoites than those positive by ELISA ($F = 8.45$; df = 1, 36; $P = 0.006$).

**Sensitivity and Specificity of Dissection versus ELISA.** Dissection and ELISA results in relation to sporozoite detection from Giemsa-stained dissection slides for the group of 352 *Anopheles* held 10 d after capture are summarized in Table 5. The sensitivity of sporozoite detection was 92.1% for dissection compared with 78.9% for ELISA. Dissection underestimated the slide sporozoite rate by 7.9% (3/38), but ELISA failed to detect CS protein in gland or thorax samples in 21.0% (8/38) of the slide positives. Specificity was 100% for dissection and 92.0% for ELISA. Overall, 45.4% (25/55) of the ELISA positives were from gland negative *Anopheles*.

## Discussion

In the development of IRMA and ELISA techniques for sporozoite detection, the ultimate objective was to provide a method for accurately determining sporozoite infection rates in field-collected *Anopheles* (Zavala et al. 1982, Burkot et al. 1984, Wirtz et al. 1985). A close agreement between dissection and ELISA infection rates was found for field-collected *Anopheles* in Papua New Guinea (Wirtz et al. 1987a), and similar concordance was observed in the Gambia using the IRMA (Collins et al. 1984). However, Esposito et al. (1986) found that IRMA infection rates were consistently higher than dissection sporozoite rates in Burkina Faso and Mali, and Boudin et al. (1988) observed the same for ELISA in Burkina Faso. An initial evaluation of dissection and ELISA techniques in western Kenya demonstrated a 78.8% higher ELISA rate and concluded that the difference was due primarily to CS protein detected from oocysts on the midgut, which could be eliminated by testing anterior portions of cut mosquitoes (J. C. Beier et al. 1987). In the current study, the 43% higher ELISA infection rate, obtained by evaluating 44,674 *Anopheles*, was not due to oocysts because tests were done only on head–thorax samples.

The *P. falciparum* ELISA, which is capable of detecting about 100 sporozoites per mosquito (Wirtz et al. 1987b), may not be sufficiently sensitive to detect sporozoites in all infected *Anopheles*. Although Pringle (1966) reported that >97% of infected Afrotropical *Anopheles* contain >500 salivary gland sporozoites, in this study 36.8% (14/38) of the gland-infected *Anopheles* contained <500 sporozoites and 26.3% (10/38) contained <100 sporozoites. Therefore, higher ELISA infection rates were more likely due to ELISA detection of CS protein (Verhave et al. 1988), which is widely disseminated throughout the mosquito body (Robert et al. 1988), than to significant dissection error.

The detection of CS protein by ELISA in 5.2% of 10,017 salivary gland samples considered negative by dissection raised the possibility of significant dissection error. Lombardi et al. (1987) also found a high rate of CS protein by IRMA in gland samples from dissection negative *Anopheles* and concluded that low sporozoite loads are missed frequently by dissection. This interpretation is inconsistent with our previous finding that the rate of ELISA identification of sporozoites from dissected glands ranges from 64.6% ($n = 99$) when <10 sporozoites are observed by dissection to 96.5% ($n = 632$) when glands contain >1,000 (M. S. Beier et

**Table 5. Sensitivity and specificity of dissection and *P. falciparum* ELISA methods for sporozoite detection, evaluated against Giemsa-stained dissection slide results for 352 field-collected *Anopheles* held 10 d after capture**

| Method | No./test category | | | | Sensitivity | Specificity |
|---|---|---|---|---|---|---|
| | Slide positive ($n = 38$) | | Slide negative ($n = 314$) | | | |
| | Positive | Negative | Positive | Negative | | |
| Dissection | 35 | 3 | 0 | 314 | 92.1% | 100.0% |
| ELISA | 30 | 8 | 25 | 289 | 78.9% | 92.0% |

al. 1988). Because at least 30% of salivary gland samples from low-grade sporozoite infections in western Kenya cannot be identified using ELISAs to the three common *Plasmodium* species, the 5.2% rate of CS protein in gland negative samples was too high to be due solely to dissection error.

Detection of CS protein by ELISA in thoraces from dissection positive *Anopheles*, after removing salivary glands, showed that ELISA activity for head–thorax body parts was not specific for sporozoites located in salivary glands. For dissection negative *Anopheles*, the rate of CS protein in thoraces (12.2%) exceeded the ELISA positive rate for glands (5.2%), perhaps because of the larger mass of tissue tested. Given the high infection rates in our study area, some of the gland negative mosquitoes with CS protein in the thorax likely represented infections where sporozoites had not yet reached the salivary glands.

Holding field-collected *Anopheles* 10 d after capture to ensure adequate sporogonic development did not affect the high rate of ELISA detection of CS protein in the salivary glands or thoraces from dissection negative *Anopheles*. The significant relationship between counted salivary gland sporozoites and ELISA absorbance values of gland and thorax samples indicated that the Giemsa-stained slide method, used as a standard for sporozoite detection, was reliable, and that the ELISA positive reactions for dissection negative glands represented CS protein not in the form of sporozoites. Failure of ELISA to detect 21% (8/38) of all sporozoite infections that were confirmed by slides was mainly due to insufficient sensitivity; six of eight mosquitoes contained <15 sporozoites. Because about 5% of salivary gland sporozoite infections in our study sites consist of *P. malariae* or *P. ovale* (M. S. Beier et al. 1988), non-*P. falciparum* sporozoites would not have been detected using only the *P. falciparum* ELISA. However, the eight gland samples that did not react in the *P. falciparum* ELISA also were negative when tested by ELISA for *P. malariae*, *P. ovale*, and *P. vivax* (J.C.B., unpublished data). Thus, microscopic detection of sporozoites on stained slides was far more sensitive than ELISA. In comparison, routine dissections did fail to detect three infections, containing 5, 9, and 16 sporozoites observed on stained slides alone. This resulted in a dissection error of 7.9%. Some low-grade sporozoite infections may have been missed even on Giemsa-stained slides, but it is unlikely that these would have been detected by ELISA (Table 4).

The 44.7% ELISA overestimate of the proportion of *Anopheles* containing salivary gland sporozoites was nearly identical to the discrepancy observed in the 20-mo longitudinal study. The error associated with ELISA infection rates was twofold: false negatives due to failure to detect low-grade sporozoite infections and false positives when sporozoites were not present in salivary glands (45.4% of the ELISA positives were from gland negative *Anopheles*). In terms of estimating vector infectivity, almost half of the ELISA positive *Anopheles* in this study did not have detectable sporozoites in the salivary glands.

Absorbance values >0.50 were observed for more than half of the ELISA positive *Anopheles* without gland sporozoites. This indicated significant levels of CS protein and not simply a statistical artifact of the cutoff method used for determining positively. Doubling the mean absorbance of negative controls yields *P. falciparum* ELISA test results with 98% consistency when samples were tested repeatedly (J. C. Beier et al. 1988). Therefore, ELISA positive reactions in the absence of gland sporozoites would not be eliminated by adjusting the absorbance threshold for positive reactions.

Our study confirms earlier findings in west Africa (Lombardi et al. 1987) that, in a large proportion of *An. gambiae* s.l., sporozoites fail to migrate to the salivary glands. We estimated that *P. falciparum* infections in about 10% of the *An. gambiae* s.l. and *An. funestus* did not result in salivary gland infections. The presence of CS protein not in the form of recognizable sporozoites in the salivary glands was likely due to *P. falciparum* infections that were aborted before sporozoites invaded the salivary glands. Detection of CS protein in sporozoite negative glands was not associated significantly with the presence of oocysts; 84.1% of *Anopheles* with *P. falciparum* oocysts contained salivary glands negative by ELISA. Further studies are required to determine why sporozoites fail to migrate to the salivary glands in a large proportion of infected Afrotropical *Anopheles*. This does not appear to be related to ingestion of CS antibodies by infected *Anopheles* (J. C. Beier et al. 1989), as suggested by Lombardi et al. (1987).

In conclusion, the immunological detection of CS protein in head–thorax samples of Afrotropical malaria vectors is significantly less accurate for determining *Anopheles* infectivity than standard dissection techniques. These findings raise important considerations for the use of ELISA or IRMA for measuring vector infectivity in malaria field studies. The accuracy of entomological inoculation rates depends upon standardized measures of the man-biting rate and the sporozoite rate, both of which are difficult to measure and are subject to error. Measures of infectivity provided by immunoassays are useful for measuring relative changes in transmission over time or for comparing relative transmission intensity in different epidemiological areas. However, entomological inoculation rates calculated using ELISA or IRMA infection rates may be overestimated because these infection rates do not represent the actual proportion of *Anopheles* with sporozoites in the salivary glands. In this respect, the use of immunoassays in malaria field studies can complicate the interpretation of malaria sporozoite transmission and when possible, results

should be standardized against traditional dissection techniques.

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